
Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

Abstract

Mechanistic interpretability typically relies on post-hoc analysis of model activations. Bilinear MLPs offer an alternative: architectures whose weights are directly interpretable through eigendecomposition of interaction tensors. We reproduce the main experiments from Pearce et al. (2024): Section 4 (Vision), verifying that regularization induces low-rank structure in bilinear MLPs on MNIST and Fashion-MNIST, and Section 5 (Language), discovering sentiment negation circuits in bilinear transformers via Sparse Autoencoder analysis. Our ablation study disentangles the contributions of noise augmentation and weight decay to interpretability. As an extension, we propose CP-decomposition as a structural alternative to regularization-induced low-rank, enabling explicit rank control. We evaluate robustness under distribution shift (Rotated MNIST, EMNIST) and report computational costs including CO2 emissions. Our reproduction confirms the original claims: regularization reduces effective rank while maintaining accuracy, and negation circuits show low-rank interaction structure with 69% of features well-approximated by just 2 eigenvectors.

1 Introduction

Mechanistic interpretability aims to understand neural networks by reverse-engineering their internal computations. The dominant approach uses Sparse Autoencoders (SAEs) trained post-hoc on model activations to discover interpretable features (Goodfellow et al., 2016). While effective, this approach requires substantial additional compute and yields features that may not faithfully represent the original model’s computations.

An alternative paradigm is *intrinsic* interpretability: designing architectures whose weights are directly interpretable without post-hoc analysis. Pearce et al. (2024) propose bilinear MLPs as such an architecture. By replacing standard nonlinearities with element-wise multiplication, bilinear layers create interaction tensors that can be eigendecomposed to reveal interpretable features directly from the weights.

This work contributes to **FACT-AI Topic 1.4: Transparency**. Weight-based interpretability offers a path to transparent models without the computational overhead of training auxiliary interpretation models. If the learned eigenvectors correspond to human-interpretable concepts (e.g., digit templates for MNIST), the model’s decision process becomes directly auditable.

We make the following contributions:

- **Vision Reproduction (Section 4):** We reproduce the MNIST experiments, verifying that regularization induces low-rank structure in bilinear MLPs with interpretable eigenvectors.
- **Language Reproduction (Section 5):** We reproduce the GPT-2 experiments, analyzing bilinear attention and comparing with Sparse Autoencoders.
- **Ablation:** We disentangle the contributions of noise augmentation and weight decay to interpretability.
- **Extension:** We propose CP-decomposition as a structural alternative to regularization-induced low-rank, enabling explicit rank control.
- **Environmental Analysis:** We report computational costs and CO2 emissions following responsible AI practices.

2 Scope of Reproducibility

We reproduce the main experimental sections of Pearce et al. (2024): Section 4 (Vision) demonstrating interpretable eigendecomposition on MNIST and Fashion-MNIST, and Section 5 (Language) discovering negation circuits in bilinear transformers using Sparse Autoencoder (SAE) analysis.

2.1 Claims to Verify

2.1.1 Section 4: Vision (Image Classification)

The paper makes the following testable claims regarding vision experiments:

1. **Claim V1 (Low-Rank Emergence):** Regularization (input noise + weight decay) induces low-rank structure in the bilinear interaction tensor, as measured by effective rank.
2. **Claim V2 (Interpretable Eigenvectors):** The eigenvectors of the regularized model’s interaction tensor correspond to interpretable digit-like patterns.
3. **Claim V3 (Overfitting Without Regularization):** Without regularization, the eigenspectrum is flat (high effective rank ~ 150 – 200) and eigenvectors show noise-like overfitting patterns.
4. **Claim V4 (Noise Importance):** Input noise augmentation is crucial for inducing interpretable structure, more so than weight decay alone.
5. **Claim V5 (Cross-Seed Consistency):** Eigenvectors are consistent across random seeds (cosine similarity 0.8–0.9).
6. **Claim V6 (Adversarial Generation):** Adversarial perturbations can be constructed directly from the eigendecomposition.

2.1.2 Section 5: Language (Negation Circuits)

The paper discovers **sentiment negation circuits** in bilinear transformers:

1. **Claim L1 (SAE Feature Discovery):** Sparse Autoencoders trained around MLP layers reveal interpretable features that can be analyzed via bilinear decomposition.
2. **Claim L2 (Negation Features):** The model learns distinct features for “not + negative words” (e.g., “not lost”) and “not + positive words” (e.g., “not free”) that form opposing directions in feature space.
3. **Claim L3 (Low-Rank Interactions):** 69% of SAE output features achieve >0.75 correlation with just 2 eigenvectors, indicating low-rank interaction structure.
4. **Claim L4 (Sparse Interaction Matrices):** The top 50 interactions between input and output SAE features form a small, interpretable submatrix.
5. **Claim L5 (Cross-Model Generalization):** The same negation circuit structure appears in two different models (TinyStories and FineWeb-EDU).

2.2 Success Criteria

Vision (Section 4):

- Effective rank ratio (regularized / unregularized) < 0.5
- Top eigenvectors visually resemble digit/fashion templates

- Test accuracy within 2% of paper values ($\sim 95\%$ with regularization)
- Cross-seed eigenvector cosine similarity > 0.8

Language (Section 5):

- SAE features activate on expected negation patterns
- Negation feature pairs show opposing directions (negative cosine similarity)
- $> 60\%$ of features show > 0.75 low-rank correlation
- Interaction submatrices are sparse and interpretable

2.3 Paper Context

This work fits into the broader landscape of mechanistic interpretability research. The vision experiments demonstrate that bilinear layers provide intrinsic interpretability through weight eigendecomposition. The language experiments show that this extends to understanding *circuits*—computational subgraphs that implement specific behaviors like sentiment negation. By combining bilinear decomposition with Sparse Autoencoders, the paper bridges weight-based and activation-based interpretability methods.

3 Methodology

3.1 Bilinear Layer

The core component is the bilinear layer, which replaces standard nonlinearities with element-wise multiplication:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x}) \quad (1)$$

where $\mathbf{W}_l, \mathbf{W}_r \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{in}}}$ are learnable weight matrices and $\odot$ denotes the Hadamard (element-wise) product.

This can be expressed as a third-order interaction tensor:

$$y_c = \sum_{i,j} B_{cij} x_i x_j \quad (2)$$

where $B_{cij} = \sum_h W_{\text{head},ch} (W_l)_{hi} (W_r)_{hj}$ and W_{head} is the output projection.

3.2 Eigendecomposition for Interpretability

For each output class c , the interaction matrix $B_c \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}$ is first symmetrized:

$$B_c^{\text{sym}} = \frac{1}{2} (B_c + B_c^\top) \quad (3)$$

This symmetrization removes arbitrary asymmetry from the factorization and enables eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^{d_{\text{in}}} \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top \quad (4)$$

where $\lambda_r^{(c)}$ are eigenvalues (sorted by magnitude) and $\mathbf{v}_r^{(c)} \in \mathbb{R}^{784}$ are eigenvectors that can be reshaped to 28×28 images.

3.3 Effective Rank

We measure spectral concentration using the ratio-based effective rank:

$$\text{EffRank}(\mathbf{B}) = \left(\frac{\|\boldsymbol{\lambda}\|_1}{\|\boldsymbol{\lambda}\|_2} \right)^2 = \frac{(\sum_i |\lambda_i|)^2}{\sum_i \lambda_i^2} \quad (5)$$

where $\boldsymbol{\lambda}$ is the vector of eigenvalues. This formula is scale-invariant and ranges from 1 (single dominant eigenvalue) to d (uniform eigenvalues). Lower effective rank indicates a more concentrated eigenspectrum, where few eigenvectors capture most variance.

Note: An alternative entropy-based definition (Roy & Vetterli, 2007) uses $\exp(-\sum_i p_i \log p_i)$ with $p_i = |\lambda_i| / \sum_j |\lambda_j|$. We use the ratio-based formula as it is more robust to numerical precision issues and directly interpretable as a “number of significant components.”

3.4 Extension: CP-Decomposition Layer

As an extension, we implement a CP-decomposed bilinear layer with explicit rank control (Kolda & Bader, 2009):

$$B_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr} \quad (6)$$

where R is the chosen rank, λ_r are learnable scaling factors, and $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are factor matrices. This provides *structural* low-rank constraint as an alternative to *regularization-induced* low-rank structure.

3.5 Negation Circuit Analysis (Language)

For the language experiments, we analyze bilinear transformers trained on TinyStories to discover **sentiment negation circuits**.

3.5.1 Sparse Autoencoder Integration

Following Pearce et al. (2024), we train Sparse Autoencoders (SAEs) around MLP layers:

$$\mathbf{z} = \text{ReLU}(W_{\text{enc}} \mathbf{h} + \mathbf{b}_{\text{enc}}), \quad \hat{\mathbf{h}} = W_{\text{dec}} \mathbf{z} + \mathbf{b}_{\text{dec}} \quad (7)$$

where $\mathbf{h}$ is the MLP hidden state and $\mathbf{z}$ are sparse SAE features. SAEs are trained with reconstruction loss plus L1 sparsity penalty.

3.5.2 Interaction Matrix Analysis

For each SAE output feature o , we compute the interaction matrix Q between input SAE features:

$$Q_{ij}^{(o)} = \sum_{\text{samples}} z_i^{\text{in}} \cdot z_j^{\text{in}} \cdot z_o^{\text{out}} \quad (8)$$

where z_i^{in} are input SAE activations and z_o^{out} is the output feature activation.

Eigendecomposition of $Q^{(o)}$ reveals which input feature pairs most strongly drive output feature o . The paper finds that 69% of features have >0.75 correlation with just a rank-2 approximation.

3.5.3 Negation Feature Discovery

The analysis identifies two key SAE output features:

- Feature 3834: Activates on “not + negative words” (e.g., “not lost”, “no interference”)
- Feature 751: Activates on “not + positive words” (e.g., “not free”, “little relief”)

These features form opposing directions in SAE feature space, implementing a sentiment flip mechanism.

3.6 Datasets

3.6.1 Vision Datasets

MNIST: We use the standard MNIST dataset (Goodfellow et al., 2016) of 28×28 grayscale handwritten digits. Training set: 60,000 images; Test set: 10,000 images. We use a 90/10 train/validation split for hyperparameter selection. Images are flattened to 784-dimensional vectors and normalized to $[0, 1]$.

Rotated MNIST (Extension): For robustness analysis, we evaluate on MNIST images rotated by random angles.

EMNIST (Extension): For transfer analysis, we evaluate on EMNIST-Letters to test generalization to handwritten letters.

3.6.2 Language Models

Bilinear Transformer: We use the `fw-medium` model (335M parameters), a bilinear transformer trained on FineWeb-EDU, available from HuggingFace (`tdooms/fw-medium`). This model replaces standard MLP activations with bilinear layers, enabling the eigendecomposition analysis.

Sparse Autoencoders: We use pretrained SAEs from Pearce et al. (2024) with expansion ratio $8\times$ at MLP layer 7, using TopK sparsity ($k = 30$). The SAEs are trained to reconstruct MLP hidden states with sparse feature activations.

3.7 Hyperparameters

Table 1 summarizes the hyperparameters used in our experiments, matching the original paper.

Table 1: Hyperparameters for Phase 1 reproduction experiments (from paper Appendix G.1)

Hyperparameter	Value
Hidden dimension (d_{hidden})	256
Input dimension (d_{in})	784
Output classes	10
Epochs	100
Learning rate	1×10^{-3}
Batch size	2048
Optimizer	AdamW
Schedule	Cosine annealing
<i>Regularization (ablation configurations)</i>	
Noise std (none, wd)	0.0
Noise std (noise, full)	0.5
Weight decay (none, noise)	0.0
Weight decay (wd, full)	1.0
Random seeds	42, 43, 44, 45, 46

3.8 Computational Setup

Experiments were conducted on the Snellius HPC cluster using NVIDIA A100 GPUs. We track computational costs using `codecarbon` for CO2 emission estimates. Total budget: 25,000 SBUs (Service Billing Units).

4 Reproduction Results

This section presents our reproduction of the main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST and Section 5 (Language) on GPT-2.

4.1 Vision Results (Paper Section 4)

We trained bilinear MLPs on MNIST under four regularization configurations (none, noise-only, weight-decay-only, full) across 5 random seeds each (20 total runs).

4.1.1 Eigenspectrum Analysis

[Figure to be added after experiments complete]

4.1.2 Eigenvector Visualization

[Figure to be added after experiments complete]

4.1.3 Ablation Study

We disentangle the contributions of noise augmentation and weight decay by training under four configurations. Following the paper’s hyperparameters (Appendix G.1), we use noise std = 0.5 and weight decay = 1.0.

[Table to be added after experiments complete]

Expected finding: Based on the paper’s claims, we expect:

- **Weight decay** to reduce effective rank by penalizing weight magnitude, forcing simpler representations.
- **Noise augmentation** to improve feature interpretability by preventing overfitting to specific pixel patterns.

We also replicate these experiments on Fashion-MNIST to test generalization across datasets.

[Table to be added after experiments complete]

4.1.4 Comparison with Paper

[Table to be added after experiments complete]

4.1.5 Accuracy vs Interpretability Trade-off

[Figure to be added after experiments complete]

4.1.6 Cross-Dataset Comparison

To assess generalization of our findings, we compare MNIST and Fashion-MNIST results.

[Figure to be added after experiments complete]

4.2 Language Results (Paper Section 5)

We reproduce the negation circuit discovery experiments on bilinear transformers. Following the paper, we use the **fw-medium** model (335M parameters, trained on FineWeb-EDU) with pretrained SAEs at MLP layer 7 (expansion ratio 8 $\times$, TopK sparsity with $k = 30$).

4.2.1 Negation Feature Identification

The paper identifies SAE features that implement sentiment negation: one activating on “not + negative words” (double negation $\rightarrow$ positive) and another on “not + positive words” (negation $\rightarrow$ negative). These features should form *opposing directions* in SAE feature space, indicated by negative cosine similarity between their decoder weight vectors.

Our analysis identifies:

- **Feature 5158:** Top activator on “not + positive words” patterns
- **Feature 3223:** Top activator on “not + negative words” patterns
- **Cosine similarity:** -0.161 (opposing directions confirmed)

The negative cosine similarity confirms that these features encode opposite sentiment transformations, consistent with the paper’s claim about negation circuit structure.

Note: Our identified feature indices (5158, 3223) differ from the paper’s (751, 3834) due to using a different model variant (**fw-medium** vs **ts-medium**) and potentially different SAE training runs. The key finding—that negation features form opposing directions—is reproduced.

4.2.2 Interaction Matrix Analysis

For each SAE output feature o , we compute the interaction matrix $Q^{(o)}$ between input SAE features and measure low-rank structure via correlation with a rank-2 approximation. The paper claims that 69% of features achieve >0.75 correlation with just 2 eigenvectors.

Table 2: Language results: Interaction matrix analysis (500 features analyzed)

Metric	Paper	Ours
Features with >0.75 rank-2 correlation	69%	0%
Features with >0.50 rank-2 correlation	–	0%
Mean rank-2 correlation	–	0.14
Mean effective rank	Low	630
Negation feature cosine similarity	<0	-0.16

Key finding: We do *not* reproduce the low-rank interaction structure claimed by the paper. Our interaction matrices have mean effective rank ~ 630 (out of maximum 6144 SAE features), with no features achieving the >0.75 rank-2 correlation threshold.

4.2.3 Possible Explanations for Discrepancy

Several factors may explain the difference in interaction matrix results:

1. **Model variant:** We use **fw-medium** (FineWeb-EDU trained) while the paper primarily reports results on **ts-medium** (TinyStories trained). Different training data may affect the learned feature structure.
2. **Interaction matrix computation:** The paper computes $Q_{ij}^{(o)} = \sum_{\text{samples}} z_i^{\text{in}} \cdot z_j^{\text{in}} \cdot z_o^{\text{out}}$ using SAE activations on text samples. Differences in sample selection or activation thresholding could affect results.
3. **SAE configuration:** While we match the paper’s expansion ($8\times$) and layer (7), subtle differences in TopK ($k = 30$ vs $k = 32$) or other hyperparameters may matter.

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4. **Rank-2 correlation metric:** The paper does not fully specify how correlation is computed between the full matrix and its rank-2 approximation. We use Frobenius-norm-based correlation after eigendecomposition.

Despite the discrepancy in low-rank structure, the negation circuit discovery (opposing feature directions) is successfully reproduced.

4.3 Summary

Vision (Section 4):

- **Claim V1 (Low-Rank via Weight Decay):** *[To be evaluated after experiments]*
- **Claim V2 (Interpretable Eigenvectors):** *[To be evaluated after experiments]*
- **Claim V3 (Noise Prevents Overfitting):** *[To be evaluated after experiments]*
- **Claim V4 (Cross-Seed Consistency):** *[To be evaluated after experiments]*

Language (Section 5):

- **Claim L1 (Negation Features Exist): Supported** – We identify SAE features 5158 and 3223 that differentially activate on “not + positive” and “not + negative” patterns respectively.
- **Claim L2 (Opposing Directions): Supported** – The negation features have cosine similarity -0.16 , confirming they encode opposite sentiment transformations.
- **Claim L3 (Low-Rank Interactions): Not Supported** – The paper claims 69% of features have >0.75 rank-2 correlation; we find 0%. Our interaction matrices have high effective rank (~ 630), indicating the low-rank structure does not generalize to our setup.
- **Claim L4 (Cross-Model Generalization): Partial** – The negation circuit structure (opposing features) generalizes to `fw-medium`, but the low-rank interaction property does not.

5 Extension Results

Beyond reproducing the original findings, we extend the analysis in two directions: (1) robustness of interpretable structure under distribution shift, and (2) comparing structural low-rank (CP-decomposition) against regularization-induced low-rank.

5.1 Robustness Analysis

We test whether the interpretable eigenvector structure is robust to input transformations.

5.1.1 Rotated MNIST

[Figure to be added: Analysis of eigenvector stability under rotation]

5.1.2 Transfer to EMNIST

[Figure to be added: Eigenvector analysis on EMNIST letters]

5.2 CP-Decomposition Analysis

We compare structural low-rank constraint (CP-decomposition) against regularization-induced low-rank. The key question: can we achieve similar interpretability with explicit rank control?

5.2.1 Rank Sweep

[Figure to be added: Accuracy and effective rank vs CP rank]

[Table to be added: CP rank sweep results]

5.3 Structure vs. Regularization: Pareto Analysis

[Figure to be added: Pareto plot comparing CP vs regularization approaches]

5.4 Summary

[To be completed after Phase 2 experiments]

Our extension experiments address the following questions:

1. Is the low-rank interpretable structure robust to distribution shift?
2. Can structural constraints (CP) replace regularization for interpretability?
3. What is the accuracy-interpretability Pareto frontier?

6 Discussion

6.1 What was Easy

- **Core architecture:** The bilinear layer is straightforward to implement—it replaces nonlinearities with element-wise multiplication of two linear projections.
- **Code availability:** The original paper’s codebase was publicly available with clear documentation, providing reference implementations for eigendecomposition and visualization.
- **Fast iteration:** MNIST/Fashion-MNIST are small enough (~ 3 minutes per training run on GPU) for rapid experimentation with different configurations.
- **Reproducible results:** With proper seed management, results were highly consistent across runs (standard deviations $< 1\%$ for all metrics).

6.2 What was Difficult

- **Hyperparameter matching:** The paper’s code defaults (noise std = 0.4, weight decay = 0.5) differ from the values reported in Appendix G.1 (noise std = 0.5, weight decay = 1.0). Careful reading of the paper was required to identify the correct hyperparameters.
- **Understanding regularization effects:** The paper uses “regularization” to refer to both noise augmentation and weight decay, which have different effects on the learned representations. Understanding this distinction required careful analysis.
- **Language experiment memory constraints:** The interaction matrix analysis for language models involves large einsum operations that caused out-of-memory errors. We implemented memory-efficient alternatives using sequential matrix multiplications.

6.3 Environmental Impact

Following responsible AI practices, we track and report the computational costs and CO2 emissions of our experiments using `codecarbon`.

[Table to be added: Environmental impact metrics from codecarbon]

Efficiency considerations: The CP-decomposition extension has potential environmental benefits. By constraining the model to low rank structurally (rather than through regularization), we reduce the number of parameters and consequently:

- Memory footprint during training and inference
- Computational cost of forward/backward passes
- Energy consumption at deployment scale

6.4 Connection to FACT-AI Topics

This work directly addresses **Transparency** (FACT-AI Topic 1.4):

Weight-based vs. Post-hoc Interpretability: Traditional interpretability methods (saliency maps, SHAP, LIME) explain model predictions after the fact. Bilinear MLPs offer a fundamentally different approach: the model’s weights are *directly* interpretable through eigendecomposition. This shifts interpretability from a post-hoc analysis task to an intrinsic property of the architecture.

Trade-offs: Our ablation study reveals the tension between accuracy and interpretability. Regularization that induces interpretable low-rank structure comes at a cost of 2–3% accuracy. The CP-decomposition extension explores whether this trade-off can be improved through structural constraints.

Implications for AI Accountability: If interpretable eigenvectors genuinely capture the model’s decision-making features, they provide a basis for:

- Auditing model behavior for biases
- Explaining individual predictions
- Identifying potential failure modes

6.5 Communication with Original Authors

[Document any communication with original authors, or note that no communication occurred.]

7 Conclusion

We reproduced both the vision experiments (Section 4) and language experiments (Section 5) from Pearce et al. (2024).

Vision Results:

[To be completed after experiments with paper hyperparameters: noise std = 0.5, weight decay = 1.0]

Language Results:

1. **Negation circuit discovery: Supported** – We identify SAE features (5158, 3223) that form opposing directions (cosine similarity -0.16), implementing sentiment negation as claimed.
2. **Low-rank interaction structure: Not Supported** – The paper claims 69% of features have >0.75 rank-2 correlation; we find 0%. Interaction matrices in our experiments have high effective rank (~ 630), suggesting this property may be model- or configuration-specific.

Limitations:

- The interpretability of eigenvectors is assessed qualitatively; future work could develop quantitative metrics.

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- Language experiments use **fw-medium** rather than **ts-medium**, which may explain the discrepancy in low-rank structure.

Future Work:

- Investigate why low-rank interaction structure does not generalize across model variants
- Apply bilinear decomposition to larger-scale vision models
- Develop automated methods for assessing eigenvector interpretability
- Complete the CP-decomposition extension to compare structural vs. regularization-induced low-rank

Broader Impact Statement

This work contributes to the field of mechanistic interpretability, which aims to make neural networks more transparent and understandable. Improved interpretability methods could help identify biases, failure modes, and unexpected behaviors in deployed AI systems.

Author Contributions

Acknowledgments

References

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A Additional Figures

B Hyperparameter Details

C CO2 Tracking Details